

Seminar On Algebraic Geometry
Lecture Notes 2026 (Draft)

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1 Sheaves

1.1 Definitions & Examples

Grothendieck's creature in algebraic geometry arises much from differential geometry. Roughly speaking, Grothendieck generalized both differential geometry and algebraic geometry, in order to provide a new geometric view with algebraic objects.

Definition 1.1. Let $(X, \mathcal{T})$ be a topology space. A **presheaf** $\mathscr{F}$ of category $\mathcal{C}$ is a contravariant functor

$$\mathscr{F} : \mathsf{Top}(X) \to \mathscr{C},$$

where $\mathrm{Top}(X) = (\mathcal{T}, \subseteq)$ is a preorder category.

In other words, if $\mathscr{F}$ is a presheaf of Ab, then for every open subset U of X, there is an abelian group $\mathscr{F}(U)$ and for every inclusion $U \subseteq V$ of open subsets of X, there is a morphism of abelian groups $\rho_{VU} = \mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$. And it satisfies

(1) $\mathscr{F}(\varnothing) = 0$; (Not necessary)

(2) $\rho_{UU} = \mathrm{id}_U : \mathscr{F}(U) \to \mathscr{F}(U)$;

(3) If $U \subseteq V \subseteq W$ are three open subsets, then $\rho_{WU} = \rho_{WV} \circ \rho_{VU}$.

$$\begin{array}{ccc} U & \longrightarrow & \mathscr{F}(U) \\ \big\downarrow & & \big\uparrow{\scriptstyle \rho_{VU}} \\ V & \longrightarrow & \mathscr{F}(V) \end{array}$$

As a matter of terminology, if : $\mathscr{F}$ is a presheaf on X, we refer to: $\mathscr{F}(U)$ as the **sections** of the presheaf $\mathscr{F}$ over the open set U, and we sometimes use the notation $\Gamma(U, \mathscr{F})$ to denote the group $\mathscr{F}(U)$. We call the maps $\rho_{VU} = \mathrm{res}_{V,U}$ **restriction maps**, and we sometimes write $s\,|_V$ instead of $\rho_{VU}(s)$, if $s \in \mathscr{F}(U)$.

We always assume that $\mathcal{C}$ be Ab, i.e., the category of abelian groups, in order to give the category of presheaves the abelian category structure.

Simply, a presheaf is just a contravariant functor $\mathscr{F} : \mathcal{D} \to \mathcal{C}$. A sheaf is a presheaf whose sections are, in a technical sense, uniquely determined by their restrictions, via "glueing together".

Definition 1.2. A presheaf $\mathscr{F}$ on a topological space X is a **sheaf** if it satisfies the following supplementary conditions:

(4) **Identity axiom:** If U is an open subset, if $\{V_i\}$ is an open covering of U, and if $s \in \mathscr{F}(U)$ is an element such that $s|_{V_i} = 0$ for all i, then $s = 0$. It is equivalent with that $s|_{V_i} = t|_{V_i} \implies s = t$.

(5) **Glueing axiom:** If U is an open subset, if $\{V_i\}$ is an open covering of U, and if we have elements $s_i \in \mathscr{F}(V_i)$ for each i, with the property that for each i, j, $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$ then there is an (by (4), unique) element $s \in \mathscr{F}(U)$ such that $s|_{V_i} = s_i$ for each i.

Lemma 1.3. Let $\mathscr{F}$ be a sheaf on a topological space X. If U and V are disjoint open subsets of X, then $\mathscr{F}(U \amalg V) \cong \mathscr{F}(U) \times \mathscr{F}(V)$.

Proof. The isomorphism is given by $s \mapsto (s|_U, s|_V)$, and the inverse is given by $(s_1, s_2) \mapsto s$, where s is glued by s_1 and s_2, since $\{U, V\}$ is naturally an open cover of $U \amalg V$. □

Example 1.4. [Ha, Exercise II.1.1] Let A be an abelian group (resp., ring), and define the **constant presheaf** $\underline{A}_{\text{pre}}$ associated to A on the topological space X to be the presheaf $U \mapsto A$ for all $U \neq \varnothing$, with restriction maps the identity. In particular, the constant presheaf $\underline{\mathbb{Z}}_{\text{pre}}$ associated to $\mathbb{Z}$ is given by $U \mapsto \mathbb{Z}$, and any section $s \in \underline{\mathbb{Z}}_{\text{pre}}(U)$ is an integer $s \in \mathbb{Z}$.

[Ha, Example II.1.0.3] Let X be a topological space, and A an abelian group (resp., ring). We define the **constant sheaf** $\underline{A}$ on X determined by A as follows.

Give A the discrete topology, and for any open set $U \in X$, let $\underline{A}(U)$ be the group (resp., ring) of all continuous maps of U into A:

$$\underline{A}(U) := C(U, A).$$

The restriction maps are the usual restrictions of maps. Since A is a discrete topological group (resp., ring), each $\underline{A}(U)$ is endowed with group (resp., ring) structure. Immediately, $\underline{A}$ is an abelian group sheaf (resp., ring) on X.

Note that for every connected open set U, $\underline{A}(U) \cong A$, whence the name "constant sheaf". To see this, let $s \in \underline{A}(U)$ be a section, then $U = \bigcup_{a \in A} s^{-1}(a)$. Since U is connected, there is only one $s^{-1}(a) = U$ and the others are empty.

Thus, we obtain a surjective homomorphism below

$$\underline{A}(U) \to A \quad \text{with} \quad s \mapsto s(x), \forall x \in U.$$

It is also injective, since $s(U) = \{0\}$ if and only if $s = 0$.

Therefore, if U is an open set whose connected components are open, then $\underline{A}(U) = \prod A$, one for each connected components of U.

Here we define sheaves of modules in order to take a more geometric viewpoint on sheaves. They are studied in [Ha, II.5 Sheaves of Modules].

Definition 1.5. A **ringed space** is a pair $(X, \mathscr{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathscr{O}_X$ on X.

Definition 1.6. Let $(X, \mathscr{O}_X)$ be a ringed space. A **sheaf of $\mathscr{O}_X$-modules** is a sheaf $\mathscr{F}$ on X, such that for each open set $U \subseteq X$, the abelian group $\mathscr{F}(U)$ is an $\mathscr{O}_X$-module, and for for each inclusion of open sets $V \subseteq U$, the restriction homomorphism $\mathscr{F}(U) \to \mathscr{F}(V)$ is compatible with the module structure via the ring homomorphism $\mathscr{O}_X(U) \to \mathscr{O}_X(V)$, that is, for any $a \in \mathscr{O}_X(U)$ and $s \in \mathscr{F}(U)$, we have $(a \cdot s)|_V = (a|_V) \cdot (s|_V)$.

Far from being abstract nonsense, sheaf theory generalizes and unifies many geometric objects. Here are some examples from geometry and analysis.

Example 1.7. Let X be a topological space. For each open subset $U \subseteq X$, let $C(U)$ denote the $\mathbb{R}$-algebra of continuous functions $U \to \mathbb{R}$, with restriction maps given by restriction of functions. Then $\mathscr{C}$, defined by $\mathscr{C}(U) = C(U)$, is a sheaf of $\mathbb{R}$-algebras on X; the glueing axiom follows from the local nature of continuity. In particular, if M is a smooth manifold, then the structure sheaf $\mathscr{O}_M$ is the sheaf of smooth functions, given by

$$\mathscr{O}_M(U) = C^\infty(U).$$

Example 1.8. Let X be a complex manifold. For each open subset $U \subseteq X$, let $\mathcal{O}(U)$ denote the $\mathbb{C}$-algebra of holomorphic functions $U \to \mathbb{C}$, with restriction maps given by restriction of functions. Then the structure sheaf $\mathscr{O}_X$, defined by $\mathscr{O}_X(U) = \mathcal{O}(U)$, is a sheaf of $\mathbb{C}$-algebras on X: if $\{U_i\}$ is an open cover of U and $f_i \in \mathcal{O}(U_i)$ agree on all overlaps $U_i \cap U_j$, then there is a unique $f \in \mathcal{O}(U)$ restricting to each f_i, since holomorphicity is a local property. This

is the holomorphic analogue of the sheaf of smooth functions on a smooth manifold.

Already here one sees why one works with sheaves rather than only global sections: on a compact connected complex manifold X, every global holomorphic function is constant, so $\mathscr{O}_X(X) \cong \mathbb{C}$, whereas the sheaf $\mathscr{O}_X$ still encodes rich local holomorphic data.

Let M be a smooth manifold. For each open subset $U \subseteq M$, let $\Omega^k(U)$ denote the $C^\infty(U)$-module of smooth k-forms on U, with restriction maps given by restriction of forms. Then Ω^k is a sheaf of C^∞_M-modules on M.

Crucially, Ω^k is not merely a sheaf of abelian groups: for $f \in C^\infty_M(U)$ and $\omega \in \Omega^k(U)$, the product $f\omega$ is defined pointwise by $(f\omega)_p = f(p)\omega_p$, and restriction is compatible with this structure, making Ω^k a sheaf of modules over the structure sheaf $\mathscr{O}_M$. In particular, $\Omega^0 = \mathscr{O}_M$, and Ω^1 is the sheaf of sections of the cotangent bundle T^*M.

Indeed, sheaves of modules provide an analogue of fibre bundles in algebraic geometry. For example, Ω^k is a $\mathscr{O}_M$-module, and is locally free of rank $\binom{n}{k}$ over $\mathscr{O}_M$, hence corresponds to the vector bundle $\bigwedge^k T^*M$, i.e., $\Omega^k = (-, \bigwedge^k T^*M)$.

There is a deep relation between sheaves of modules with bundles. For example, in differential geometry, the Swan-Serre theorem states that: for X a smooth manifold with $\mathbb{R}$-algebra of smooth functions $C^\infty(X)$, there is an equivalence of categories between that of finite rank smooth vector bundles over X and finitely generated projective modules over $C^\infty(X)$. See [Ne, Chapter 12] for more details.

Example 1.9. Let $p : Y \to X$ be a continuous map. For each open subset $U \subseteq X$, define

$$\Gamma(Y/X)(U) = \{s : U \to Y \mid p \circ s = \mathrm{id}_U,\ s \text{ continuous}\},$$

with restriction maps given by $s \mapsto s|_V$. Then $\Gamma(Y/X)$ is a sheaf on X, called the *sheaf of sections* of p.

If $p : Y \to X$ is a vector bundle, then $\Gamma(Y/X)(U)$ is a $C^\infty(U)$-module, and $\Gamma(Y/X)$ is a sheaf of C^∞_X-modules, locally free of rank equal to the rank of the bundle. Indeed, the name "section" comes from such sections in differential geometry.

Example 1.10. Let M be a smooth manifold, for any open subset $U \subseteq X$, define $C_c^\infty(U)$ be the $\mathbb{R}$-vector space of smooth functions from U to $\mathbb{R}$ with compact support. Then C_c forms a presheaf, which is not a sheaf in general.

For example, let $M = \mathbb{R}$, φ be a bump function such that $\mathrm{supp}(\varphi) = [-1/3, 1/3]$, and $\varphi(0) = 1$. Let $f(x) = \sum_{n\in\mathbb{Z}} \varphi(x-n)$. Then $\mathrm{f}(\varphi) = \bigcup_{n\in\mathbb{Z}}[n - 1/3, n + 1/3]$, hence $f \notin C_c(\mathbb{R})$. Now let $U_n := (n - 1/2, n + 2/3)$ be some open subsets covering $\mathbb{R}$. In each U_n, $\mathrm{supp}(f\,|_{U_n}) = [n - 1/3, n + 1/3]$ which is compact. However, the global section glueing by $f\,|_{U_n}$ is identically f, which is not in $C_c^\infty(\mathbb{R})$.

An interesting result is, as the dual topological space of $C_c^\infty(U)$, the space of distributions $\mathcal{D}'(U)$ forms a sheaf. For a brief introduction to this, see [WkDi, 4.5].

Now we come back to classical algebraic geometry. In [Ha, I.3 Morphisms], we define the ring of regular functions on an algebraic variety V, denoted by $\mathscr{O}(V)$.

Example 1.11. [Ha, Example II.1.0.1] Let V be an algebraic variety over the field k. For each open set $U \subseteq V$, let $\mathscr{O}_V(U) := \mathscr{O}(U)$ be the ring of regular functions from U to k. Then $\mathscr{O}_V$ is a presheaf of rings on X.

$\mathscr{O}_V$ is a sheaf, which is called the *sheaf of regular functions* on X. This is because the regularity of a function is defined locally.

Now we define the morphisms between presheaves on the same space.

Definition 1.12. For a presheaf $\mathscr{F}$ is a contravariant functor $\mathscr{F} : \mathsf{Top}(X) \to \mathsf{Ab}$, it forms a functor category $[\mathsf{Top}(X)^{\mathrm{op}}, \mathsf{Ab}]$. We define the category of presheaves as $\mathsf{PSh}(X) = [\mathsf{Top}(X)^{\mathrm{op}}, \mathsf{Ab}]$. In general, we define $\mathsf{PSh}(X, \mathcal{C}) = [\mathsf{Top}(X)^{\mathrm{op}}, \mathcal{C}]$, for any category $\mathcal{C}$.

Thus, the morphism $\varphi : \mathscr{F} \to \mathscr{G}$ is a natural transformation, which consists of morphisms of abelian groups $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ for each open set U, such that whenever $V \subseteq U$ is an inclusion, the diagram

$$\begin{array}{ccc} \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ \downarrow{\scriptstyle \rho_{UV}} & & \downarrow{\scriptstyle \rho_{UV}} \\ \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

commutes. Or we can simply say that $\varphi(s\,|_V) = \varphi(s)\,|_V$, for any section $s \in \mathscr{F}(U)$.

We can simply define the category of sheaves on X to be the full subcategory $\mathsf{Sh}(X) := \mathsf{Sh}(X, \mathsf{Ab})$ of $\mathsf{PSh}(X)$.

Definition 1.13. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves. We define the presheaf kernel of φ, presheaf cokernel of φ, and presheaf image of φ to be the presheaves given by $U \mapsto \ker \varphi(U)$, $U \mapsto \operatorname{coker} \varphi(U)$, and $U \mapsto \operatorname{im} \varphi(U)$ respectively.

Indeed, PSh is an abelian category. We denote the cokernel of φ in PSh by $\operatorname{coker}_{\mathsf{PSh}} \varphi$ and the image of φ in PSh by $\operatorname{im}_{\mathsf{PSh}} \varphi$, since they are not sheaves in general. However, the kernel of φ in PSh is a sheaf, by the lemma below.

Lemma 1.14. If $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves, then the presheaf kernel of φ, $\ker \varphi$ is also a sheaf.

Proof. Notice that $(\ker \varphi)(U) \subseteq \mathscr{F}(U)$ hence we can use the sheaf axiom on $\mathscr{F}$. □

1.2 Stalks & Germs

Before we investigate the category of sheaves and operations on sheaves, we first introduce the stalks of sheaves.

Definition 1.15 (Stalks)**.** If $\mathscr{F}$ is a presheaf on X, and if P is a point of X, we define the *stalk* $\mathscr{F}_P$ of $\mathscr{F}$ at P to be the direct limit (colimit) of the groups $\mathscr{F}(U)$ for all open sets U containing P, via the restriction maps ρ, as

$$\mathscr{F}_P := \varinjlim \mathscr{F}(U).$$

The direct limit means that, for any open neighborhood U of P, there exists a homomorphism $\rho_U : \mathscr{F}(U) \to \mathscr{F}_P$, such that, for compatible homomorphisms $\lambda_U : \mathscr{F}(U) \to A$, there exists a unique homomorphism $\mathscr{F}_P \to A$.

Elements in $\mathscr{F}_P$ are called *germs* of sections of $\mathscr{F}$ at the point P.

In fact, the index category $\mathsf{Top}_P(X)$ with all open neighborhood U of P is a *filtered category*, since for any open neighborhood U and V of P, $U \cap V$ is also an open neighborhood of P. By [Va, 1.4 D], the germ in $\mathscr{F}_P$ has the form of $\langle U, s\rangle$ while U is the open neighborhood of P and $s \in \mathscr{F}(U)$ under the equivalence relation that

$$\langle U, s\rangle = \langle V, t\rangle \iff \exists \text{ an open neighborhood } W \subseteq U \cap V \text{ of } P \text{ s.t. } s|_W = t|_W .$$

In fact, taking stalks gives a functor $\mathsf{Sh}(X) \to \mathsf{Ab}$, which is exact and left-adjoint.

Example 1.16. Let X be a complex manifold and let $\mathscr{O}_X$ be the sheaf of holomorphic functions on X. If $\dim_{\mathbb{C}} X = n$ and $z_1, \ldots, z_n$ are local holomorphic coordinates centered at $P \in X$, then the germ of a holomorphic function is uniquely determined by its Taylor expansion at P. Hence

$$\mathscr{O}_{X,P} \cong \mathbb{C}\{z_1, \ldots, z_n\},$$

the ring of convergent power series in n variables. This stalk is a local ring, whose maximal ideal consists of germs vanishing at P.

Example 1.17. The name of germs came from analytic continuation. See [Ah, Chapter 8].

A *function element* is a pair (D, f) consisting of an open disk $D \subset \mathbb{C}$ and a holomorphic function f on D. Two function elements (D, f) and (E, g) define the same *germ* at $z_0 \in D \cap E$ if $f = g$ in a neighborhood of z_0. The set of all such germs, denoted $\mathcal{O}$, carries a natural topology: for each function element (D, f), the collection $\{[f]_z : z \in D\}$ is a basic open neighborhood of each of its germs. The projection

$$\pi : \mathcal{O} \to \mathbb{C}, \qquad [f]_z \mapsto z,$$

is then a local homeomorphism, because the map $z \mapsto [f]_z$ is a homeomorphism from D onto $\{[f]_z : z \in D\}$ with inverse π. Thus $\mathcal{O}$ is a one-dimensional complex manifold, i.e. a Riemann surface, and π is holomorphic. Its connected components are the Riemann surfaces of the *global analytic functions* obtained by analytic continuation: continuing a germ $[f]_a$ along a curve γ from a to b amounts to lifting γ to $\mathcal{O}$, so the continuation becomes a single-valued holomorphic function on the component containing $[f]_a$.

Example 1.18. More generally, let X be a topological space. An *étale space* over X is a continuous map $\pi : \mathcal{F} \to X$ which is a local homeomorphism. The fiber $\mathcal{F}_x = \pi^{-1}(x)$ is the *stalk* at x. If each stalk carries the structure of an abelian group and the group operations are continuous, this is the old-fashioned definition of a *sheaf of abelian groups* on X; the sections over an open set U are the continuous maps $s : U \to \mathcal{F}$ with $\pi \circ s = \mathrm{id}_U$. The modern sheaf $\mathscr{F}$ associated with an étale space is obtained by taking all sections over open subsets.

1.3 Local Nature of Sheaves I

Lemma 1.19. For every $P \in U$, there is a map $\mathscr{F}(U) \to \mathscr{F}_P$, sending a section $s \in \mathscr{F}(U)$ to the equivalence class of (U, s), denoted s_P.

If $\mathscr{F}$ is a sheaf, a section $s \in \mathscr{F}(U)$ of $\mathscr{F}$ over U is determined by its images in the stalks $\mathscr{F}_x$ for all $x \in U$, i.e. $s = 0$ if and only if $s_P = 0$, for all $P \in U$. Equivalently, $s = t \in \mathscr{F}(U)$ if and only if $s_P = t_P \in \mathscr{F}_P$, for all $P \in U$.

Proof. For any germ (s_P, U_P), such that $S_P = 0$, we can obtain the zero section by glueing s_P, since U_P forms an open cover of X. By the identity axiom, any section glueing by s_P must be zero. □

Proposition 1.20. [Va, 2.4 N], [Ha, Exercise II.1.3] Consider the morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of sheaves on the topology space X. Then, the conditions below are equivalent:

(1) φ is a monomorphism.

(2) φ is injective on the level of stalks: $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is injective for all $P \in X$.

(3) $\ker \varphi = 0$.

(4) φ is injective on the level of open sets : $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ is a monomorphism of Ab, thus injective and $\ker \varphi(U) = (0)$, for all open set $U \subseteq X$.

We say that φ is *injective* if it satisfies the above equivalent conditions.

Proof. We prove this by (2) $\Longrightarrow$ (1) $\Longrightarrow$ (3) $\Longrightarrow$ (4) $\Longrightarrow$ (2).

(2) $\Longrightarrow$ (1) : By 1.19.

(1) $\Longrightarrow$ (3) : By the definition of $\ker \varphi$ we could obtain an inclusion morphism $i : \ker \varphi \to \mathscr{F}$ and a zero morphism $0 : \ker \varphi \to \mathscr{F}$ sending $(\ker \varphi)(U) \mapsto (0)$ inducing the commutative diagram below

$$\ker \varphi \underset{0}{\overset{i}{\rightrightarrows}} \mathscr{F} \xrightarrow{\varphi} \mathscr{G}.$$

Since φ is a monomorphism, i must be 0, which means that $\ker \varphi = 0$.

(3) $\Longrightarrow$ (4) : Notice that $\ker(\varphi(U)) = (\ker \varphi)(U)$.

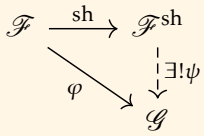
(4) $\Longrightarrow$ (2) : If $\ker \varphi(U) = (0)$ for all open sets $U \subseteq X$, then for any $P \in X$, and germs $s_P \in \ker \varphi_P$, there exists an open neighborhood U of P, such that $s_P = \langle U, s \rangle$ for some $s \in \ker \varphi(U)$. By the assumption, $s = 0$, hence $s_P = 0$.

□

Since $\mathrm{im}_{\mathsf{PSh}}$ is not a sheaf in general, the case of epimorphisms is more complicated. We need to introduce the sheafification of presheaves, in order to obtain the image of a morphism of sheaves.

1.4 Sheafification

Proposition 1.21. [Ha, Proposition II.1.2] Given a presheaf $\mathscr{F}$, there is a sheaf $\mathscr{F}^{\mathrm{sh}}$ and a morphism $\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ with the following universal property: for any sheaf $\mathscr{G}$ and any morphism of presheaves $\varphi : \mathscr{F} \to \mathscr{G}$ (where $\mathscr{G}$ is regarded as a presheaf), there is a unique morphism of sheaves $\psi : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ such that $\varphi = \psi \circ \mathrm{sh}$.



Furthermore the pair $(\mathscr{F}^{\mathrm{sh}}, \mathrm{sh})$ is unique up to a unique isomorphism. We call $\mathscr{F}^{\mathrm{sh}}$ *the sheaf associated to the presheaf* $\mathscr{F}$, or *the sheafification of* $\mathscr{F}$.

Proof. We construct the sheafification $\mathscr{F}^{\mathrm{sh}}$ as follows. For any open subset U, let $\mathscr{F}^{\mathrm{sh}}(U)$ be the set of functions

$$s : U \longrightarrow \coprod_{P \in U} \mathscr{F}_P$$

such that

(1) for each $P \in U$, $s(P) \in \mathscr{F}_P$, and

(2) for each $P \in U$, there is an open neighborhood $V \subseteq U$ of P, and an element $t \in \mathscr{F}(V)$, such that for all $Q \in V$, the germ t_Q of t at Q is equal to $s(Q)$.

By construction, sh is naturally defined by sending $s \in \mathscr{F}(U)$ to the function $P \mapsto s_P$. The restriction maps are defined by restricting the domain of these functions.

We now verify that $\mathscr{F}^{\mathrm{sh}}$ is a sheaf. For the identity axiom, let $U \subseteq X$ be open, let $\{U_i\}$ be an open cover of U, and let $s \in \mathscr{F}^{\mathrm{sh}}(U)$ satisfy $s|_{U_i} = 0$ for all i. Since the restriction maps are restrictions of functions, $s|_{U_i}(P) = s(P)$ for every $P \in U_i$. Given $P \in U$, choose i with $P \in U_i$. Then

$$s(P) = s|_{U_i}(P) = 0.$$

Hence $s = 0$.

For the glueing axiom, suppose we are given sections $s_i \in \mathscr{F}^{\mathrm{sh}}(U_i)$ such that

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

for all i, j. Define $s : U \to \coprod_{P \in U} \mathscr{F}_P$ by $s(P) = s_i(P)$ whenever $P \in U_i$. This is well defined by the compatibility on overlaps. To see that $s \in \mathscr{F}^{\mathrm{sh}}(U)$, fix $P \in U$ and choose i with $P \in U_i$. Since $s_i \in \mathscr{F}^{\mathrm{sh}}(U_i)$, there is an open neighborhood $V \subseteq U_i$ of P and an element $t \in \mathscr{F}(V)$ such that $t_Q = s_i(Q)$ for all $Q \in V$. Then $t_Q = s(Q)$ for all $Q \in V$, so the local condition holds at P. Thus $s \in \mathscr{F}^{\mathrm{sh}}(U)$, and $s|_{U_i} = s_i$ for every i, because the two functions agree at each point of U_i. Therefore $\mathscr{F}^{\mathrm{sh}}$ is a sheaf.

Now we prove the universal property. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves, with $\mathscr{G}$ a sheaf. Let $s \in \mathscr{F}^{\mathrm{sh}}(U)$. For each $P \in U$, by the construction of $\mathscr{F}^{\mathrm{sh}}$, there is an open neighborhood $V_P \subseteq U$ of P and an element $t^P \in \mathscr{F}(V_P)$ such that

$$s(Q) = t^P_Q \qquad \text{for all } Q \in V_P.$$

We claim that for any $P, Q \in U$, the sections $\varphi(t^P)$ and $\varphi(t^Q)$ agree on $V_P \cap V_Q$. Indeed, for every $R \in V_P \cap V_Q$, we have

$$\varphi_R\big(t^P_R\big) = \varphi_R\big(s(R)\big) = \varphi_R\big(t^Q_R\big).$$

Thus the germs of $\varphi(t^P)$ and $\varphi(t^Q)$ at every point of $V_P \cap V_Q$ coincide. Since $\mathscr{G}$ is a sheaf, Lemma 1.19 implies

$$\varphi(t^P)|_{V_P \cap V_Q} = \varphi(t^Q)|_{V_P \cap V_Q}.$$

Therefore, by the glueing axiom for the sheaf $\mathscr{G}$, there exists a unique section $\psi(s) \in \mathscr{G}(U)$ such that

$$\psi(s)|_{V_P} = \varphi(t^P) \qquad \text{for all } P \in U.$$

The section $\psi(s)$ is independent of the choices of V_P and t^P: if W_P, u^P is another choice, then on $V_P \cap W_P$ the sections $\varphi(t^P)$ and $\varphi(u^P)$ have the same germs at every point, hence are equal because $\mathscr{G}$ is a sheaf; uniqueness in the glueing axiom then gives the same global section.

It is immediate from the local description that ψ is compatible with restriction maps, hence ψ is a morphism of sheaves. Moreover, for $t \in \mathscr{F}(U)$, the section $\psi(\mathrm{sh}(t))$ agrees locally with $\varphi(t)$, so $\psi \circ \mathrm{sh} = \varphi$.

Finally, suppose that $\psi' : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ is another morphism of sheaves such that $\psi' \circ \mathrm{sh} = \varphi$. For $s \in \mathscr{F}^{\mathrm{sh}}(U)$ and $P \in U$, using the same local data V_P, t^P, we have

$$\psi'(s)|_{V_P} = \psi'(\mathrm{sh}(t^P))|_{V_P} = \varphi(t^P) = \psi(s)|_{V_P}.$$

Since $\mathscr{G}$ is a sheaf, this implies $\psi'(s) = \psi(s)$. Hence $\psi' = \psi$. The uniqueness of $(\mathscr{F}^{\mathrm{sh}}, \mathrm{sh})$ up to a unique isomorphism follows formally from the universal property. □

Lemma 1.22. [Va, 2.4 M] The sheafification $\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ induces an isomorphism on stalks:

$$\mathrm{sh}_P : \mathscr{F}_P \longrightarrow \mathscr{F}_P^{\mathrm{sh}}, \qquad \forall P \in X.$$

Conversely, let $\mathscr{F}$ be a presheaf, let $\mathscr{G}$ be a sheaf, and let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves. If the induced map $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is an isomorphism for all $P \in X$, then the induced morphism $\varphi^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ is an isomorphism.

Proof. First we prove that sh_P is injective. Let $\alpha \in \mathscr{F}_P$, and choose a representative $s \in \mathscr{F}(U)$ of α. Suppose that $\mathrm{sh}_P(\alpha) = 0$. Then $\mathrm{sh}(s)_P = 0$ in $\mathscr{F}_P^{\mathrm{sh}}$. Hence there is an open neighborhood $V \subseteq U$ of P such that

$$\mathrm{sh}(s)|_V = 0 \qquad \text{in } \mathscr{F}^{\mathrm{sh}}(V).$$

For every $Q \in V$, we have

$$\mathrm{sh}(s)(Q) = s_Q.$$

Therefore $s_Q = 0$ for every $Q \in V$. In particular, taking $Q = P$, we obtain $s_P = 0$ in $\mathscr{F}_P$, that is, $\alpha = 0$. Thus sh_P is injective.

Next we prove that sh_P is surjective. Let $\beta \in \mathscr{F}_P^{\mathrm{sh}}$, and choose a representative $\sigma \in \mathscr{F}^{\mathrm{sh}}(U)$ of β. By the construction of $\mathscr{F}^{\mathrm{sh}}$, after possibly shrinking U to an open neighborhood V of P, there exists $t \in \mathscr{F}(V)$ such that

$$t_Q = \sigma(Q) \qquad \text{for all } Q \in V.$$

Hence $\mathrm{sh}(t)|_V = \sigma|_V$, and therefore

$$\mathrm{sh}_P(t_P) = \mathrm{sh}(t)_P = \sigma_P = \beta.$$

Thus sh_P is surjective. Consequently sh_P is an isomorphism for every $P \in X$.

For the converse, the universal property of sheafification gives a unique morphism $\varphi^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ such that $\varphi^{\mathrm{sh}} \circ \mathrm{sh} = \varphi$. On stalks, since sh_P is an isomorphism, we obtain

$$\varphi_P^{\mathrm{sh}} = \varphi_P \circ \mathrm{sh}_P^{-1}.$$

By assumption φ_P is an isomorphism for every $P \in X$. Hence φ_P^{sh} is an isomorphism

for every $P \in X$. By Proposition 1.30, φ^{sh} is an isomorphism. □

Proposition 1.23. [Va, 2.4 H] Let $\mathscr{F},\mathscr{G} \in \mathrm{PSh}(X)$ be a presheaf of X, let $\mathscr{F}^{\mathrm{sh}},\mathscr{G}^{\mathrm{sh}} \in \mathrm{Sh}(X)$ be the sheafification of them. If there is a morphism $\varphi : \mathscr{F} \to \mathscr{G}$, then there exists a (unique) natural induced morphism of sheaves $\varphi^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}^{\mathrm{sh}}$, such that the following diagram commutes.

$$\begin{array}{ccc} \mathscr{F} & \xrightarrow{\varphi} & \mathscr{G} \\ \downarrow{\scriptstyle \mathrm{sh}} & & \downarrow{\scriptstyle \mathrm{sh}} \\ \mathscr{F}^{\mathrm{sh}} & \xrightarrow{\varphi^{\mathrm{sh}}} & \mathscr{G}^{\mathrm{sh}} \end{array}$$

Proof. This follows from the universal property of sheafification. □

Remark 1.24. Indeed, Proposition 1.23 shows that the sheafification is a functor $\mathsf{sh} : \mathsf{PSh} \to \mathsf{Sh}$. Moreover, it is left-adjoint to the inclusion functor $\mathsf{Sh} \to \mathsf{PSh}$, i.e., for any presheaf $\mathscr{F}$ and sheaf $\mathscr{G}$,

$$\mathrm{Hom}_{\mathsf{Sh}}(\mathscr{F}^{\mathrm{sh}},\mathscr{G}) \cong \mathrm{Hom}_{\mathsf{PSh}}(\mathscr{F},\mathscr{G}).$$

Example 1.25. Let A be an abelian group, and X be a topological space. We have defined the constant presheaf $\underline{A}_{\mathrm{pre}}$ on X, and the constant sheaf $\underline{A}$ on X in Example 1.4. We claim that $\underline{A}$ is the sheafification of $\underline{A}_{\mathrm{pre}}$.

Indeed, the natural morphism $\underline{A}_{\mathrm{pre}} \to \underline{A}$ is the identity on stalks: both stalks are A, since a germ of a locally constant function is constant on a small neighborhood. Hence this morphism is an isomorphism on all stalks, and therefore the induced morphism $\underline{A}^{\mathrm{sh}}_{\mathrm{pre}} \to \underline{A}$ is an isomorphism on stalks by Lemma 1.22. It follows that $\underline{A}$ is the sheafification of $\underline{A}_{\mathrm{pre}}$.

One can check that each sheaves of abelian groups on X is a $\underline{\mathbb{Z}}$-sheaves on X by definition. We will prove this by $\mathscr{H}om$ in the next time.

Now we can define the cokernel and image of a morphism of sheaves, hence giving a complete description of the abelian category of sheaves.

Definition 1.26. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves.

- We define the kernel of φ to be the presheaf kernel of φ, $\ker \varphi$.
- We define the cokernel of φ to be the sheafification of the presheaf cokernel of φ, $\mathrm{coker}\, \varphi := (\mathrm{coker}_{\mathsf{PSh}}\, \varphi)^{\mathrm{sh}}$.
- We define the image of φ to be the sheafification of the presheaf image

of φ, $\operatorname{im}\varphi := (\operatorname{im}_{\mathsf{PSh}}\varphi)^{\mathrm{sh}}$.

Corollary 1.27. [Ha, Exercise II.1.4] Let $\varphi : \mathscr{F} \to \mathscr{G}$ be an injective morphism of sheaves. Then the induced map $\varphi^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}^{\mathrm{sh}}$ of sheafifications is injective. Therefore, we can view $\operatorname{im}\varphi$ as a natural subsheaf of $\mathscr{G}$.

Proof. We check this on the level of stalks. Notice that $\varphi_P^{\mathrm{sh}} = \mathrm{sh}_P \circ \varphi_P \circ \mathrm{sh}_P^{-1}$ since sh_P is an isomorphism, by 1.22. Therefore, φ_P^{sh} is injective for all $P \in X$, since φ_P is injective for all $P \in X$. □

1.5 Local Nature of Sheaves II

By giving the definition of image in the category of sheaves, we can now give an analogue of Proposition 1.20 for epimorphisms.

Proposition 1.28. [Va, 2.4 O], [Ha, Ex 1.3] Consider the morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of sheaves on the topology space X. Then, the conditions below are equivalent:

(1) φ is an epimorphism.

(2) φ is surjective on the level of stalks: $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is surjective for all $P \in X$.

(3) $\operatorname{im}\varphi = \mathscr{G}$.

(4) For every open set $U \subseteq X$, and for every section $s \in \mathscr{G}(U)$, there is a covering $\{U_i\}$ of U, and sections $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s|_{U_i}$ for all i.

We say that φ is *surjective* if it satisfies the above equivalent conditions.

Proof. We prove this by (2) $\implies$ (1) $\implies$ (3) $\implies$ (4) $\implies$ (2). The proof of (2) $\implies$ (1) and (1) $\implies$ (3) is similar to Proposition 1.20.

(3) $\implies$ (4) : Let U be an open set, and $s \in \mathscr{G}(U)$ be a section. Since $\operatorname{im}\varphi = \mathscr{G}$, for each $P \in U$, there exists an open neighborhood $V_P \subseteq U$ of P, and a section $t^P \in \mathscr{F}(V_P)$ such that $\varphi(t^P) = s|_{V_P}$, because of the construction of the sheafification. Then $\{V_P\}_{P\in U}$ is an open cover of U, and the sections t^P satisfy $\varphi(t^P) = s|_{V_P}$ for all $P \in U$.

(4) $\implies$ (2) : For every point $P \in X$, and every germ $\langle s, U\rangle \in \mathscr{G}_P$, there is a covering $\{U_i\}$ of U and sections $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s|_{U_i}$, by assumption.

Choose an open neighborhood U_i of P, then $\langle s, U\rangle = \langle s, U_i\rangle = \langle \varphi(t_i), U_i\rangle$ in $\mathscr{G}_P$. This proves φ_P is surjective. □

Corollary 1.29. [Ha, Exercise II.1.2] A sequence of sheaves

$$\cdots \to \mathscr{F}^{i-1} \xrightarrow{\varphi^{i-1}} \mathscr{F}^{i} \xrightarrow{\varphi^{i}} \mathscr{F}^{i+1} \to \cdots$$

is exact if and only if the induced sequence of stalks

$$\cdots \to \mathscr{F}_P^{i-1} \xrightarrow{\varphi_P^{i-1}} \mathscr{F}_P^{i} \xrightarrow{\varphi_P^{i}} \mathscr{F}_P^{i+1} \to \cdots$$

for all $P \in X$.

Proposition 1.30. [Ha, Proposition II.1.1] Let Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves on a topological space X. Then φ is an isomorphism if and only if the induced map on the stalk $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is an isomorphism for every $P \in X$.

Proof. By Proposition 1.20 and Proposition 1.28. □

Proposition 1.31. [Ha, Exercise II.1.2] For any morphism of sheaves $\varphi : \mathscr{F} \to \mathscr{G}$, for each point $P \in X$, $(\ker \varphi)_P = \ker(\varphi_P)$ and $(\operatorname{im} \varphi)_P = \operatorname{im}(\varphi_P)$.

Proof. Left to readers. □

1.6 Abelian Category Of Sheaves

Let X be a topological space.

Definition 1.32. Let $\mathscr{F}'$ be a subsheaf of a sheaf $\mathscr{F}$, i.e. there is an injective morphism $i : \mathscr{F}' \to \mathscr{F}$ (we can view $\mathscr{F}'(U)$ as a subsheaf of $\mathscr{F}(U)$ for each open subset U). We define the quotient sheaf $\mathscr{F}/\mathscr{F}'$ to be the sheafification of the presheaf $U \mapsto \mathscr{F}(U)/\mathscr{F}'(U)$. For any $P \in X$, we have $(\mathscr{F}/\mathscr{F}')_P = \mathscr{F}_P/\mathscr{F}'_P$.

Corollary 1.33. [Ha, Exercise II.1.7] Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves. Then $\operatorname{im} \varphi \cong \mathscr{F}/\ker \varphi$, and $\operatorname{coker} \varphi \cong \mathscr{G}/\operatorname{im} \varphi$.

Proposition 1.34 (Direct sum)**.** [Ha, Exercise II.1.9]

Proposition 1.35 (Direct limit)**.** [Ha, Exercise II.1.10, 1.11]

Proposition 1.36 (Inverse limit)**.** [Ha, Exercise II.1.12]

1.7 Operations on Sheaves

1.7.1 Global Section Functor

Recall that we sometimes write $\Gamma(U,\mathscr{F}) := \mathscr{F}(U)$, for any open subset $U \subseteq X$ and any sheaf $\mathscr{F}$ on X.

Proposition 1.37. [Ha, Ex 1.8] Suppose $U \subseteq X$ is an open subset, and $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is an exact sequence of sheaves of abelian groups. Then

$$0 \to \Gamma(U,\mathscr{F}) \to \Gamma(U,\mathscr{G}) \to \Gamma(U,\mathscr{H})$$

is exact. I.e., $\Gamma(U,-)$ is a left exact functor $\mathrm{Sh}(X) \to \mathrm{Ab}$.

Proof. We can easily check this on the level of stalks, since for any morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of sheaves, and for any $P \in U$, we have $\ker\varphi_P = (\ker\varphi)_P$ and $\operatorname{im}\varphi_P = (\operatorname{im}\varphi)_P$. □

However, $\Gamma(U,-)$ is not exact in general, especially in algebraic geometry: a surjective morphism of sheaves need not induce a surjection on sections. The following two examples illustrate that differential topology is "soft," whereas the Zariski topology is "hard."

Lemma 1.38. Let M be a smooth (paracompact) manifold and let $\{U_i\}_{i\in I}$ be an open cover of M. Then there exists a locally finite partition of unity subordinate to $\{U_i\}$: a family $\{\rho_i\}_{i\in I}$ of smooth functions $\rho_i \in C^\infty(M)$ such that $0 \le \rho_i \le 1$, $\operatorname{supp}\rho_i \subseteq U_i$ is compact, the family $\{\operatorname{supp}\rho_i\}_{i\in I}$ is locally finite, and $\sum_{i\in I}\rho_i = 1$ on M (the sum is locally finite, hence well defined).

Example 1.39. Let M be a smooth manifold, $\mathscr{C}^\infty_M$ the sheaf of smooth functions on M, and $\mathscr{C}^\infty_{M,c}$ the presheaf of compactly supported smooth functions. The inclusion $\mathscr{C}^\infty_{M,c} \hookrightarrow \mathscr{C}^\infty_M$ induces an isomorphism

$$(\mathscr{C}^\infty_{M,c})^{\mathrm{sh}} \xrightarrow{\sim} \mathscr{C}^\infty_M$$

on sheafification. Indeed, for each $f \in C^\infty(M)$ and each point, choose a lo-

cally finite open cover $\{U_i\}_{i\in I}$ of M by relatively compact open subsets and a partition of unity $\{\rho_i\}_{i\in I}$ subordinate to it. Then each $f_i := \rho_i f$ is compactly supported and only finitely many f_i are nonzero near each point, so $f = \sum_i f_i$ with $\sum_i f_i \in \Gamma(M, \mathscr{C}^\infty_{M,c})$. Thus every smooth function is locally a sum of compactly supported smooth functions, and the sheafification of $\mathscr{C}^\infty_{M,c}$ is $\mathscr{C}^\infty_M$.

Proposition 1.40. Let M be a smooth manifold and $\mathscr{O}_M := \mathscr{C}^\infty_M$ the sheaf of smooth functions. Then the global section functor

$$\Gamma(M, -) : \mathscr{O}_M\text{-Mod} \longrightarrow \mathsf{Ab}$$

is exact.

Proof. Since $\Gamma(M,-)$ is left exact, it suffices to show that every surjective morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of $\mathscr{O}_M$-modules induces a surjection on global sections. Let $g \in \Gamma(M, \mathscr{G})$. As φ is an epimorphism of sheaves, there exist an open cover $\{U_i\}_{i\in I}$ of M and local lifts $f_i \in \mathscr{F}(U_i)$ with $\varphi(f_i) = g|_{U_i}$. Choose a partition of unity $\{\rho_i\}_{i\in I}$ subordinate to $\{U_i\}_{i\in I}$. Since $\mathscr{F}$ is an $\mathscr{O}_M$-module, the section $\rho_i f_i \in \mathscr{F}(U_i)$ vanishes near the boundary of U_i, hence extends by zero to a global section $\widetilde{\rho_i f_i} \in \Gamma(M, \mathscr{F})$. By local finiteness the sum $f := \sum_{i\in I} \widetilde{\rho_i f_i}$ is a well-defined global section of $\mathscr{F}$, and

$$\varphi(f) = \sum_{i\in I} \rho_i \, \varphi(f_i) = \sum_{i\in I} \rho_i \, g|_{U_i} = g.$$

Hence φ is surjective on global sections, and $\Gamma(M,-)$ is exact. □

Example 1.41. [Va, Example 2.4.10] Let $X = \mathbb{C}$ with the classical topology, and define $\mathscr{O}_X$ to be the sheaf of holomorphic functions, and $\mathscr{O}^*_X$ to be the sheaf of invertible holomorphic functions. Then we have an exact sequence

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathscr{O}_X \xrightarrow{\exp} \mathscr{O}^*_X \longrightarrow 1,$$

where $\underline{\mathbb{Z}}$ is the constant sheaf associated to $\mathbb{Z}$.

Indeed, we can check this on stalks. Without loss of generality, consider $P = 0/$ Then $\underline{\mathbb{Z}}_0 = \mathbb{Z}$, and $\mathscr{O}_{X,0}$ is the ring of convergent Taylor series at origin, and $\mathscr{O}^*_{X,0}$ is the ring of functions in $\mathscr{O}_{X,0}$ which doesn't vanish at 0. The image of $\times 2\pi i$ on stalks is $2\pi i\mathbb{Z}$ as constant functions, and the kernel of exp is just $2\pi i\mathbb{Z}$ by Euler's equation. To show that the expotential map is surjective on stalks, we only need to notice that for any holomorphic function $f \in \mathscr{O}^*_{X,0}$, there is an open neighborhood U of the origin such that f has no roots in U.

Thus, $\ln f$ is a holomorphic function in U, which gives the preimage of f in $\mathscr{O}_{X,0}$.

Let $U = \mathbb{C} \setminus \{0\}$ be the punctured complex plane. Suppose that there is some $f \in \mathscr{O}_X(U)$, such that $\exp f = \mathrm{id}_U \in \mathscr{O}_X^*(U)$, i.e., $\exp(f(z)) = z$ on U. Then $f'(z)z = f'(z)e^{f(z)} = 1$, for $z \in U$, and we have $f'(z) = 1/z$. Now we calculate the integral at $\gamma : |z| = 1$:

$$\oint_\gamma f'(z)\,\mathrm{d}z = \oint_\gamma \frac{1}{z}\,\mathrm{d}z = 2\pi i \neq 0.$$

However, by Cauchy's Integral Theorem, since γ is a closed curve containing in U, a holomorphic open domain of $f'(z)$, the integral must be zero. Therefore, such f does not exist, and $\exp(U)$ is not surjective.

1.7.2 Hom Sheaf

Proposition 1.42. [Ha, Exercise II.1.15] Let $\mathscr{F},\mathscr{G}$ be sheaves of abelian groups on X. For any open set $U \subseteq X$, the set $\mathscr{H}om(\mathscr{F}|_U,\mathscr{G}|_U)$ is an abelian group, and $\mathscr{H}om(\mathscr{F},\mathscr{G})$ is a sheaf of abelian groups on X, by $U \mapsto \mathrm{Hom}(\mathscr{F}|_U,\mathscr{G}|_U)$. It is called the *sheaf of local morphisms* of $\mathscr{F}$ into $\mathscr{G}$, "sheaf hom" for short, and is denoted $\mathrm{Hom}(\mathscr{F},\mathscr{G})$.

Proof. For each open $U \subseteq X$, the set $\mathrm{Hom}(\mathscr{F}|_U,\mathscr{G}|_U)$ becomes an abelian group under pointwise addition: for morphisms φ,ψ and any open $V \subseteq U$, $s \in \mathscr{F}(V)$, put

$$(\varphi+\psi)(V)(s) = \varphi(V)(s) + \psi(V)(s),$$

and define $(-\varphi)(V)(s) = -\varphi(V)(s)$, the zero morphism sending every section to 0. The group axioms are inherited from the abelian groups $\mathscr{G}(V)$, so $U \mapsto \mathrm{Hom}(\mathscr{F}|_U,\mathscr{G}|_U)$ is a presheaf of abelian groups.

It remains to check the sheaf axioms. Let $U \subseteq X$ be open and $\{U_i\}_{i\in I}$ an open cover of U. *Identity:* if $\varphi \in \mathrm{Hom}(\mathscr{F}|_U,\mathscr{G}|_U)$ satisfies $\varphi|_{U_i} = 0$ for all i, then for every open $V \subseteq U$ and $s \in \mathscr{F}(V)$ the sections $\varphi(V)(s)|_{V\cap U_i} = 0$ for all i; since $\mathscr{G}$ is a sheaf and $\{V \cap U_i\}_{i\in I}$ covers V, we get $\varphi(V)(s) = 0$, hence $\varphi = 0$. *Gluing:* suppose $\varphi_i \in \mathrm{Hom}(\mathscr{F}|_{U_i},\mathscr{G}|_{U_i})$ satisfy $\varphi_i|_{U_i\cap U_j} = \varphi_j|_{U_i\cap U_j}$ for all i,j. For open $V \subseteq U$ and $s \in \mathscr{F}(V)$, the sections $\varphi_i(V \cap U_i)(s|_{V\cap U_i}) \in \mathscr{G}(V \cap U_i)$ agree on the overlaps $V \cap U_i \cap U_j$; by the sheaf property of $\mathscr{G}$ they glue to a unique section $\varphi(V)(s) \in \mathscr{G}(V)$. The assignments $V \mapsto \varphi(V)$ define a morphism of sheaves $\mathscr{F}|_U \to \mathscr{G}|_U$, and $\varphi|_{U_i} = \varphi_i$ by construction. Hence the presheaf is a sheaf of abelian groups. □

Proposition 1.43. [Va, Exercise 2.6.I] Suppose $\mathscr{F}$ is a sheaf of abelian groups on a topological space X. Then $\mathscr{H}om(\mathscr{F},\cdot)$ is a left-exact functor $\mathrm{Sh}(X) \to \mathrm{Sh}(X)$, a nd $\mathscr{H}om(\cdot,\mathscr{F})$ is a left-exact contravariant functor $\mathrm{Sh}(X) \to \mathrm{Sh}(X)$, as we define above.

Proof. Since the Hom sheaf is defined on sections by $\mathscr{H}om(\mathscr{F},\mathscr{G})(U) = \mathrm{Hom}(\mathscr{F}|_U, \mathscr{G}|_U)$, and exactness of a sequence of sheaves is checked on every open set (restriction to an open subset preserves exactness), it suffices to prove the corresponding exactness of Hom in $\mathrm{Sh}(U)$ for each open U.

Let $0 \to \mathscr{G}' \xrightarrow{\alpha} \mathscr{G} \xrightarrow{\beta} \mathscr{G}''$ be exact. We show that

$$0 \to \mathrm{Hom}(\mathscr{F},\mathscr{G}') \xrightarrow{\alpha_*} \mathrm{Hom}(\mathscr{F},\mathscr{G}) \xrightarrow{\beta_*} \mathrm{Hom}(\mathscr{F},\mathscr{G}'')$$

is exact. If $\alpha \circ \varphi = 0$, then $\varphi = 0$ since α is injective; hence α_* is injective. As $\beta \circ \alpha = 0$, we have $\beta_* \circ \alpha_* = 0$. Conversely, if $\beta \circ \varphi = 0$, then φ factors through $\ker \beta = \mathrm{im}\, \alpha \cong \mathscr{G}'$ (using that α is injective), so $\varphi = \alpha \circ \psi$ for a unique $\psi \in \mathrm{Hom}(\mathscr{F},\mathscr{G}')$. Thus $\ker \beta_* = \mathrm{im}\, \alpha_*$, proving left-exactness of $\mathscr{H}om(\mathscr{F},\cdot)$. □

1.8 Pushforward & Inverse Image

Definition 1.44. Let $f : X \to Y$ be a continuous map of topological spaces.

- For any sheaf $\mathscr{F}$ on X, we define the *direct image (pushforward)* sheaf $f_*\mathscr{F}$ on Y by
$$f_*\mathscr{F}(V) := \mathscr{F}(f^{-1}V),$$
for any open set $V \subseteq Y$.
- For any sheaf $\mathscr{G}$ on Y, we define a presheaf $f^+\mathscr{G}$ on X by
$$f^+\mathscr{G}(U) = \varinjlim_{V \supseteq f(U)} \mathscr{G}(V),$$
for any open set $U \subseteq X$. And we define *inverse image* sheaf $f^{-1}\mathscr{G}$ to be the sheafification of $f^+\mathscr{G}$ given above.

Then $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ and $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ are two functors between sheaves on different topological spaces.

It's easy to verify that the pushforward $f_*\mathscr{F}$ is still a sheaf, since for any open subset $V \subseteq Y$, $f^{-1}V$ is an open subset of X. However, let $U \subseteq X$ be an open subset, it's not in general $f(V)$ is still open. Therefore, we take an approximation by

the larger open sets $V \supseteq f(U)$. More precisely, since the inclusion determines a reversed restriction map, we define the inverse image to be the limit of such $\mathscr{G}(V)$, with arrows $\mathscr{G}(V) \to f^{-1}\mathscr{G}(U)$.

If $x \in X$ is a point, and $i : \{x\} \hookrightarrow X$ is the inclusion map, then the stalk can be viewed as $\mathscr{F}_x = i^{-1}\mathscr{F}$ for every presheaf $\mathscr{F}$ on X.

Lemma 1.45. [Va, Exercise 2.2.I, 2.7.C] Suppose $f : X \to Y$ is a continuous map, and $\mathscr{F}$ is a sheaf on X, $\mathscr{G}$ is a sheaf on Y. If $f(p) = q$, then we have

$$(f_*\mathscr{F})_q \cong \mathscr{F}_p, \text{ and } \mathscr{G}_q \cong (f^{-1}\mathscr{G})_p.$$

Proof. We prove this statement by the definition of colimit. On the one hand, for each open neighborhood U of q, $f^{-1}U$ is an open neighborhood of p, thus there is a morphism $\mathscr{F}_p \to \varinjlim_{q\in U} \mathscr{F}(f^{-1}U) = (f_*\mathscr{F})_q$; on the other hand, for each open neighborhood V of p, $V \subseteq f^{-1}f(V)$ and $q \in f(V)$, which shows that $\mathscr{F}_p \cong (f_*\mathscr{F})_q$, since the limit is given by the inclusions of open subsets.

The analogue of inverse image is given by the equation

$$(f^{-1}\mathscr{G})_p = (f^+\mathscr{G})_p = \varinjlim_{p\in U} \varinjlim_{f(U)\subseteq V} \mathscr{G}(V) = \varinjlim_{q\in V} \mathscr{G}(V),$$

where the last equation is because for each open neighborhood V of q, $q \in f^{-1}(V)$ and $ff^{-1}(V) \subseteq V$. □

Proposition 1.46. [Ha, Exercise II.1.18] Let $f : X \to Y$ be a continuous map of topological space. Then $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ is left-adjoint to $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$. Precisely, there is a natural isomorphism

$$\tau_{\mathscr{F}\mathscr{G}} : \mathrm{Hom}_X(f^{-1}\mathscr{G}, \mathscr{F}) \to \mathrm{Hom}_Y(\mathscr{G}, f_*\mathscr{F})$$

for any sheaf $\mathscr{F} \in \mathrm{Sh}(X)$ and $\mathscr{G} \in \mathrm{Sh}(Y)$.

Proof. We prove the adjointness by giving the unit and counit. Notice that for open subset $V \subseteq Y$, there is a natural morphism

$$\eta_{\mathscr{G}}(V) : \mathscr{G}(V) \to f_*f^+\mathscr{G}(V) = \varinjlim_{V'\supseteq f(f^{-1}V)} \mathscr{G}(V') = \varinjlim_{V'\supseteq V\cap f(X)} \mathscr{G}(V'),$$

since $V \supseteq V \cap f(X)$. On the other side, for open subset $U \subseteq X$, there is also a natural morphism

$$\varepsilon_{\mathscr{F}}(U) : f^+f_*\mathscr{F}(U) = \varinjlim_{U'\supseteq f(U)} \mathscr{F}(f^{-1}U') \to \mathscr{F}(U),$$

since U is contained in each $f^{-1}U'$. □

Definition 1.47. If Z is a subset of X, regarded as a topological subspace with the induced topology , if $i : Z \to X$ is the inclusion map, and if $\mathscr{F}$ is a sheaf on X, then we call $i^{-1}\mathscr{F}$ the *restriction* of $\mathscr{F}$ to Z, and we often denote it by $\mathscr{F}|_Z$. If U is an open subset of X, then $\mathscr{F}|_U$ is naturally defined by $\mathscr{F}|_U(V) = \mathscr{F}(V)$ for any open subset $V \subseteq U$.

Corollary 1.48. Let Z be a subset of X, $\mathscr{F}$ be a sheaf on X. Then $(\mathscr{F}|_Z)_P = \mathscr{F}_P$ for any point $P \in Z$.

Example 1.49 (Skyscraper)**.** [Ha, Exercise II.1.17]

2 Locally Ringed Spaces

Lemma 2.1. Let $(A, \mathfrak{m}_A)$ and $(B, \mathfrak{m}_B)$ be two local rings, and $\varphi : A \to B$ be a homomorphism, then the following statements are equivalent:

(a) $f^{-1}(\mathfrak{m}_B) = \mathfrak{m}_A$,

(b) $f(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$.

If f satifies the equivalent condition above, we say that f is a **local homomorphism**.

Proof. (a) $\implies$ (b) is because $f(\mathfrak{m}_A) = ff^{-1}(\mathfrak{m}_B) = \mathfrak{m}_B \cap f(A) \subseteq \mathfrak{m}_B$. Suppose $f(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$, then $\mathfrak{m}_A \subseteq f^{-1}f\mathfrak{m}_A \subseteq f^{-1}(\mathfrak{m}_B)$. Since $\mathfrak{m}_A$ is maximal, and $f^{-1}(\mathfrak{m}_B)$ is a prime ideal, we have $f^{-1}(\mathfrak{m}_B) = \mathfrak{m}_A$, this proves (b) $\implies$ (a). □

Definition 2.2. A **ringed space** is a pair $(X, \mathscr{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathscr{O}_X$ on X.

A **morphism of ringed spaces** from $(X, \mathscr{O}_X)$ to $(Y, \mathscr{O}_Y)$ is a pair $(f, f^\sharp)$ of a continuous map $f : X \to Y$ and a morphism of sheaves $\mathscr{O}_Y \to f_*\mathscr{O}_X$ on Y.

We say the ringed space $(X, \mathscr{O}_X)$ is a **locally ringed space**, if for each point $P \in X$, the stalk $\mathscr{O}_{X,P}$ is a local ring, with a unique maximal ideal $\mathfrak{m}_{X,P}$.

A **morphism of locally ringed spaces** is a morphism $(f, f^\sharp)$ of ringed

spaces, such that for each point $P \in X$, the induced natural map

$$f_P^\sharp : \mathscr{O}_{Y,f(P)} = \varinjlim_{f(P)\in V} \mathscr{O}_Y(V) \to \varinjlim_{f(P)\in V} \mathscr{O}_X(f^{-1}V) \to \mathscr{O}_{X,P}$$

is a local homomorphism.

Notice that, by Lemma 1.45, the induced map on stalks is given by

$$f_P^\sharp : \mathscr{O}_{Y,f(P)} \to (f_*\mathscr{O}_X)_{f(P)} \cong \mathscr{O}_{X,P}.$$

Next lemma show the equivalent definition of the local induced map.

Here is a geometric view, to view a section $f \in \mathscr{O}_X(U)$ as a function. It takes values at $P \in X$ by $g(P) := g_P + \mathfrak{m}_P \in \mathscr{O}_{X,P}/\mathfrak{m}_{X,P}$. So we can restate the condition of local homomorphism by the following lemma.

Lemma 2.3. [E&H] Let $(f, f^\sharp) : X \to Y$ be a morphism of ringed spaces, where X and Y are locally ringed spaces. Then the following statements are equivalent:

(a) f is a morphism of locally ringed spaces, i.e., the induced map $f_P^\sharp : \mathscr{O}_{Y,f(P)} \to \mathscr{O}_{X,P}$ is a local homomorphism for any $P \in X$.

(b) For any point $P \in X$ and any open neighborhood V of $f(P)$ in Y, a section $s \in \mathscr{O}_Y(V)$ vanishes at $f(P)$, i.e., $s_{f(P)} \in \mathfrak{m}_{Y,f(P)}$ if and only if the section $f^\sharp(s) \in f_*\mathscr{O}_X(V) = \mathscr{O}_X(f^{-1}V)$ vanishes at P, i.e., $f^\sharp(s)_P \in \mathfrak{m}_{X,P}$.

Proof. (a) $\implies$ (b): Let $s \in \mathscr{O}_Y(V)$ be a section. Suppose that $s_{f(P)} \in \mathfrak{m}_{Y,f(P)}$. Then $(f^\sharp(s))_P = f_P^\sharp(s_{f(P)}) \subseteq \mathfrak{m}_{X,P}$, by Lemma 2.1. Conversely, suppose that $f^\sharp(s)_P \in \mathfrak{m}_{X,P}$. Then $s_{f(P)} \in f_P^{\sharp,-1}(f^\sharp(s)_P) = \mathfrak{m}_{Y,f(P)}$.

(b) $\implies$ (a): Now consider a germ $\langle s, V\rangle$, where $s \in \mathscr{O}_Y(V)$ and V is an open neighborhood of $f(P)$. Suppose that $\langle s, V\rangle \in \mathfrak{m}_{Y,f(P)}$. Then $f^\sharp(s)_P = f_P^\sharp(\langle s, V\rangle) = \langle f^\sharp(s), f^{-1}V\rangle = f^\sharp(s)_P \in \mathfrak{m}_{X,P}$. This proves that $f_P^\sharp(\mathfrak{m}_{Y,f(P)}) \subseteq \mathfrak{m}_{X,P}$, thus $f^\sharp$ is a local homomorphism. □

Example 2.4. Let $(X, \mathscr{C}_X)$ be a ringed space, where X is a topological space and $\mathscr{C}_X$ denotes the sheaf of continuous maps from X to $\mathbb{R}$ (or $\mathbb{C}$). Let $f : X \to Y$ be a continuous map, then there is a pullback

$$f^\sharp : \mathscr{C}_Y(U) \to f_*\mathscr{C}_X(U) = \mathscr{C}_X(f^{-1}U), \quad g \mapsto g \circ f$$

for any open subset $U \subseteq Y$, which gives the morphism of sheaves $f^\sharp : \mathscr{C}_Y \to f_*\mathscr{C}_X$.

Now consider the stalks. Since there continuous function $g : X \to \mathbb{R}$ can be defined locally, $\mathscr{C}_{X,P}$ is simply a $\mathbb{R}$-algebra of functions continuous at $P \in X$. $\mathscr{C}_{X,P}$ is a local ring, with a unique maximal ideal $\mathfrak{m}_{X,P} = \{h \in \mathscr{C}_{X,P} : h(P) \neq 0\}$ (because if f does not vanish at P, then there is an open neighborhood of P such that f does not vanish on it), which is the kernel of the value map $\mathfrak{m}_{X,P} \to \mathbb{R}$ given by $h \mapsto h(P)$. If $g \in \mathscr{C}_{Y,f(P)}$ is a continuous function at $f(P)$, we have $f^\sharp(g)(f(P)) = g \circ f(P) = 0$ if and only if $g(f(P)) = 0$. Therefore, $f_P^\sharp$ is a local homomorphism, by 2.3.

Example 2.5. Let $(M, \mathscr{C}_M^\infty)$ be a ringed space, where M is a smooth manifold (as a topological manifold), $\mathscr{C}_M^\infty$ is the ring of smooth functions on M. Let $(N, \mathscr{C}_N^\infty)$ be another ringed space of smooth manifold, $f : M \to N$ a continuous map. Then we can also define the pullback map $f^\sharp : \mathscr{C}_N^\infty \to f_*\mathscr{C}_M$.

Indeed, [Wed, Proposition 4.18] shows that: $f : M \to N$ is a smooth map, if and only if the pullback map $f^\sharp$ factors through $f_*\mathscr{C}_M^\infty$, i.e., the pullback of each smooth function on N is a smooth function on M. (In [Wed], the author provides a view to construct a differential manifold totally as a ringed space.) Therefore, $f^\sharp : \mathscr{C}_N^\infty \to f_*\mathscr{C}_M^\infty$ is well-defined, and the induced map $f_P^\sharp : \mathscr{C}_{Y,f(P)} \to f_*\mathscr{C}_{X,P}$ is a local homomorphism by viewing the similar arguments with the continuous version above.

Similarly, complex manifold is also a locally ringed space.

Indeed, the equivalent definition of manifolds motivate the definition of morphisms between algebraic varieties: a continuous map $f : X \to Y$ between varieties is a morphism if it pull back a regular function on a $V \subseteq Y$ to a regular function on $f^{-1}(V) \subseteq X$.

Example 2.6. Let $(X, \mathscr{O}_X)$ be a variety, with the regular function sheaf as the structure sheaf. By [Ha, Theorem 3.2, 3.4], the stalks $\mathscr{O}_{X,P}$ is a local ring (localization of the coordinate ring at the maximal ideal corresponding to P). The morphism $f : X \to Y$ of varieties is a continuous map such that $f^* : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is well-defined by sending a regular function $s \in \mathscr{O}_Y(V)$ to the pullback $s \circ f \in \mathscr{O}_X(f^{-1}V)$.

Moreover, since for each maximal ideal $\mathfrak{m}_{X,P}$ in the local ring $\mathscr{O}_{X,P}$, we have $\mathscr{O}_{X,P}/\mathfrak{m}_{X,P} \cong k$ via the valuation map $f \mapsto f(P)$. It's easy to verify that the induced map $f_P^\sharp : \mathscr{O}_{Y,f(P)} \to \mathscr{O}_{X,P}$ is a local homomorphism.

2.1 Sheaves Of Modules: An Introduction

2.2 Base Sheaves

This section is based on [E&H, Proposition I-12], [Va, 2.5 Recovering sheaves from a "sheaf on a base"] and [Sta, 6.30 Bases and sheaves].

Definition 2.7. Let X be a topological space. Let $\mathcal{B}$ be a basis for X. We say that a collection of groups $\mathscr{F}(U)$ for open sets $U \in \mathcal{B}$ and maps $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ for $V \subseteq U$ form a *$\mathcal{B}$-sheaf* if they satisfy the sheaf axiom with respect to inclusions of basic open sets in basic open sets and coverings of basic open sets by basic open sets:

- **Presheaf axiom:** $\mathscr{F}$ is a contravarient functor from the category of $\mathcal{B}$ with morphisms of inclusions, to the category of Ab.
- **Sheaf axiom:** For any $U \in \mathcal{B}$ and an open cover $\{U_i\}_{i\in I} \subseteq \mathcal{B}$ of U, if there is a collection of sections $s_i \in \mathscr{F}(U_i)$, such that for any covers U_i and U_j, and any basic open sets $V \in \mathcal{B}$ such that $V \subseteq U_i \cap U_j$, $s_i|_V = s_j|_V$, then there is a unique section $s \in \mathscr{F}(U)$, such that $s|_{U_i} = s_i$.

A morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of $\mathcal{B}$-sheaves is a set of morphisms $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$, for any $U \in \mathcal{B}$, such that $\varphi(s|_V) = \varphi(s)|_V$, if $V \subseteq U$ and $s \in \mathscr{F}(U)$.

Lemma 2.8. Let $\mathcal{B}$ be a base of open sets for X, $\mathscr{F}$ be a $\mathcal{B}$-sheaf. For any open set $U \subseteq X$, define $\mathscr{F}(U)$ as the inverse limit of the sets $\mathscr{F}'(V)$, where V runs over basic open sets contained in U:

$$\mathscr{F}'(U) = \varprojlim_{V\subseteq U, V\in\mathcal{B}} \mathscr{F}(V).$$

Then

- $\mathscr{F}'(U)$ is the set of families $(f_V)_{V\subseteq U, V\in\mathcal{B}} \in \prod_{V\subseteq U, V\in\mathcal{B}} \mathscr{F}(V)$ such that $f_V|_W = f_W$, whenever $W \subseteq V \subseteq U$ with $W, V \in \mathcal{B}$.
- $\mathscr{F}'(V) = \mathscr{F}(V)$ for all $V \in \mathcal{B}$.
- $\mathscr{F}'$ is a sheaf on X.

Proposition 2.9. [E&H, Proposition I-12] Let $\mathcal{B}$ be a base of open sets for X.

(1) Every $\mathcal{B}$-sheaf $\mathscr{F}$ on X extends uniquely to a sheaf on X, denoted by $\mathscr{F}^{\mathrm{ext}} \in \mathsf{Sh}(X)$.

(2) Every morphism of $\mathcal{B}$-sheaves $\overline{\varphi} : \mathscr{F} \to \mathscr{G}$ extends uniquely to a morphism $\varphi : \mathscr{F}^{\text{ext}} \to \mathscr{G}^{\text{ext}}$ of sheaves on X, such that $\varphi(U) = \overline{\varphi}(U)$, for all $U \in \mathcal{B}$.

The extension above forms a functor $\text{ext} : \text{Sh}(\mathcal{B}) \to \text{Sh}(X)$, from the category of $\mathcal{B}$-sheaf to the category of sheaves on X.

2.3 Gluing Sheaves And Locally Ringed Spaces

Proposition 2.10. [Ha, Exercise II.1.22] Let X be a topological space, let $\mathcal{U} = \{U_i\}$ be an open cover of X, and suppose we are given for each i a sheaf $\mathscr{F}_i$ on U_i, and for each i, j, an isomorphism $\varphi_{ij} : \mathscr{F}_i|_{U_i \cap U_j} \to \mathscr{F}_j|_{U_i \cap U_j}$, such that

(1) for each i, $\varphi_{ii} = \text{id}$, and

(2) for each i, j, k, $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_i \cap U_j \cap U_k$.

Then there exists a unique sheaf $\mathscr{F}$ on X, together with isomorphisms $\psi_i : \mathscr{F}|_{U_i} \to \mathscr{F}_i$ such that for each i, j, $\psi_j = \varphi_{ij} \circ \psi_i$ on $U_i \cap U_j$. We say that $\mathscr{F}$ is obtained by *glueing* the sheaves $\mathscr{F}_i$ via the isomorphisms φ_{ij}.

Proposition 2.11. [Ha, Exercise II.2.12] and [Sta, Lemma 26.14.1] Let $\{X_i\}$ be a family of locally ringed spaces. For each $i \neq j$, suppose given an open subspace $U_{ij} \subseteq X_i$. Suppose also given for each $i \neq j$ an isomorphism of locally ringed spaces $\varphi_{ij} : U_{ij} \to U_{ji}$ satisfying the cocycle condition:

(1) for each i, $\varphi_{ij} = \varphi_{ji}^{-1}$, and

(2) for each i, j, k, $\varphi_{ij}(U_{ij} \cap U_{ik}) = U_{ji} \cap U_{jk}$, and $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_{ij} \cap U_{ik}$.

Then there is a locally ringed space X, together with morphisms $\psi_i : X_i \to X$, such that

(1) ψ_i is an isomorphism of X_i onto an open subspace of X,

(2) the $\psi_i(X_i)$ cover X,

(3) $\psi_i(U_{ij}) = \psi_i(X_i) \cap \psi_j(X_j)$ and

(4) $\psi_i = \psi_j \circ \varphi_{ij}$ on U_{ij}.

We say that X is obtained by *glueing* the locally ringed space X_i along the isomorphisms φ_{ij}.

Moreover, if each X_i is a scheme, then X is a scheme.

3 Schemes

In our notes, a ring will always mean a commutative ring with identity.

3.1 Affine Schemes

3.1.1 Spec A as a topological space

Definition 3.1. Let A be a ring. We define $\operatorname{Spec} A$ to be a set of all prime ideals of A, and is called *spectrum* or *spectra* of ring A.

Let T be a subset of A, we define the *vanishing set* of T to be

$$V(T) := \{\mathfrak{p} \in \operatorname{Spec} A : T \subseteq \mathfrak{p}\}.$$

It's easy to see that $V(T) = V(\mathfrak{a})$, where $\mathfrak{a} = (T)$ is the ideal generated by T.

Given a subset $S \subseteq \operatorname{Spec} A$, we define $I(S)$ to be the set of functions vanishing on S, i.e.,

$$I(S) := \bigcap_{\mathfrak{p} \in S} \mathfrak{p} \subseteq A.$$

We call the next lemma the *Scheinnullstellensatz*, following the terminology in [A&K]. The term is a compound of two German words: "Nullstellensatz" for Hilbert's Nullstellensatz, and "Schein" for "formal", since Hilbert's Nullstellensatz states that

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{a} \subseteq \mathfrak{m}, \mathfrak{m} \in \operatorname{Spm} A} \mathfrak{m},$$

where A is a finitely generated k-algebra over an algebraically closed field k, and $\operatorname{Spm} A$ denotes the set of maximal ideals of A.

Lemma 3.2 (Scheinnullstellensatz)**.** Let A be a ring, and $\mathfrak{a}$ be an ideal of A. Then

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{a} \subseteq \mathfrak{p}, \mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

(By convention, the empty intersection is taken to be the whole space A.)

In other words, $\sqrt{\mathfrak{a}} = IV(\mathfrak{a}) := \bigcap_{\mathfrak{p} \in V(\mathfrak{a})} \mathfrak{p}$.

Proof. "$\subseteq$: " If $f \in \sqrt{\mathfrak{a}}$, then there exists $n \in \mathbb{N}$, such that $f^n \in \mathfrak{a} \subseteq \mathfrak{p}$, for each $\mathfrak{p} \in V(\mathfrak{a})$. Since $\mathfrak{p}$ is a prime ideal, we have $f \in \mathfrak{p}$, for each $\mathfrak{p} \in V(\mathfrak{a})$.

"$\supseteq$: " If $f \notin \sqrt{\mathfrak{a}}$, then $S = \{1, f, f^2, \cdots\}$ is a multiplicative set in $A \setminus \mathfrak{a}$. Thus $\mathfrak{a}A_f$ is an ideal in the local ring A_f, which is contained in a maximal ideal $\mathfrak{m}$ of A_f.

Consider the preimage $\mathfrak{p}$ of $\mathfrak{m}$ via the localization map, then $\mathfrak{a} \subseteq \mathfrak{p}$. But $f \notin \mathfrak{p}$, a contradiction. □

Lemma 3.3 (Zariski topology)**.** [Ha, Lemma II.2.1]

(a) If $\mathfrak{a}$ and $\mathfrak{b}$ are ideals of A, then $V(\mathfrak{ab}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$.

(b) If $\{\mathfrak{a}\}_i$ is any ideals of A, then $V(\sum_i \mathfrak{a}_i) = \bigcap_i V(\mathfrak{a}_i)$.

(c) If $\mathfrak{a}$ and $\mathfrak{b}$ are ideals of A, then $V(\mathfrak{a}) \subseteq V(\mathfrak{b})$ if and only if $\sqrt{\mathfrak{a}} \supseteq \sqrt{\mathfrak{b}}$.

So the sets $V(\mathfrak{a})$ form the closed sets of a topology on $\operatorname{Spec} A$, which is called the *Zariski topology*.

Proof. (a) Since $\mathfrak{ab} \subseteq \mathfrak{a}$, we have $V(\mathfrak{ab}) \supseteq V(\mathfrak{a})$. For the converse, let $\mathfrak{ab} \subseteq \mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal. Suppose that $a \in \mathfrak{p} \setminus \mathfrak{a}$ and $b \in \mathfrak{p} \setminus \mathfrak{b}$, then $ab \in \mathfrak{ab} \subseteq \mathfrak{p}$ but $ab \notin \mathfrak{p}$, a contradiction. Therefore, $\mathfrak{p}$ contains either $\mathfrak{a}$ or $\mathfrak{b}$.

(b) Since $\mathfrak{a}_i \subseteq \sum \mathfrak{a}_i$, we have $V(\sum \mathfrak{a}_i) \subseteq \bigcap V(\mathfrak{a}_i)$. For the converse, if $\mathfrak{p} \in \bigcap V(\mathfrak{a}_i)$, then $\mathfrak{a}_i \subseteq \mathfrak{p}$ thus $\sum \mathfrak{a}_i \subseteq \mathfrak{p}$.

(c) By Lemma 3.2.

□

We can simply have an analogue of [Ha, Proposition I.1.2] in spectrum of a ring.

Corollary 3.4. [Va, Theorem 3.7.1]

(a) If $T_1 \subseteq T_2$ are subsets of A, then $V(T_1) \supseteq V(T_2)$.

(b) If $Y_1 \subseteq Y_2$ are subsets of $\operatorname{Spec} A$, then $I(Y_1) \supseteq I(Y_2)$.

(c) For any two subsets $Y_1, Y_2 \subseteq \operatorname{Spec} A$, we have $I(Y_1 \cup Y_2) = I(Y_1) \cap I(Y_2)$.

(d) For any ideal $\mathfrak{a} \subseteq A$, $IV(\mathfrak{a}) = \sqrt{\mathfrak{a}}$.

(e) For any subset $Y \subseteq \operatorname{Spec} A$, $VI(Y) = \overline{Y}$, the closure of Y.

Therefore, $V(\cdot)$ and $I(\cdot)$ give an inclusion-reversing bijection between closed subsets of $\operatorname{Spec} A$ and radical ideals of A.

Proof. We only need to prove (e). Let $I(Y) = \mathfrak{a}$ and $\overline{Y} = V(\mathfrak{b})$. If $\mathfrak{c} \subseteq \mathfrak{a}$, and $V(\mathfrak{c}) \supseteq V(\mathfrak{a})$, then $\sqrt{\mathfrak{c}} = IV(\mathfrak{c}) \subseteq IV(\mathfrak{a}) = \mathfrak{a}$. Similarly, we have $\sqrt{\mathfrak{b}} = IV(\mathfrak{b}) = I(\overline{Y}) \subseteq I(Y) = \mathfrak{a}$. Since $Y \subseteq VI(Y)$, $\overline{Y} \subseteq VI(Y) = V(\mathfrak{a})$, then $\sqrt{\mathfrak{b}} = I(\overline{Y}) \supseteq IV(\mathfrak{a}) = \mathfrak{a}$. Thus, $\mathfrak{a} = \sqrt{\mathfrak{b}}$ and $VI(Y) = V(\mathfrak{a}) = V(\sqrt{\mathfrak{b}}) = V(\mathfrak{b}) = \overline{Y}$. □

Now we give a geometric interpretation of the relation between the ring A and the topological space $\operatorname{Spec} A$. A prime ideal $\mathfrak{p}$ in A when considered as a point in $\operatorname{Spec} A$ will be denoted by $[\mathfrak{p}]$. Elements $f \in A$ will be called **functions** on $\operatorname{Spec} A$, and the value of a function f at a point $[\mathfrak{p}]$ will be $f + \mathfrak{p}$. In this way, we call the fraction field of $A/\mathfrak{p}$ the **residue field** of $[\mathfrak{p}]$, denoted by $\kappa([\mathfrak{p}])$.

$$f : \operatorname{Spec} A \to \coprod \kappa([\mathfrak{p}]), \qquad \mathfrak{p} \mapsto f + \mathfrak{p}.$$

Hence the closed subset $V(\mathfrak{a}) = \bigcap_{f \in \mathfrak{a}} V(f)$ can be thought of as the zero locus of the functions in $\mathfrak{a}$.

Now we construct a topological basis for the Zariski topology on $\operatorname{Spec} A$. In the geometric language, a basic open set $D(f)$ is the support of the function f.

Definition 3.5. For any element $f \in A$, we define the *basic open set* $D(f)$ to be

$$D(f) := \operatorname{Spec} A \setminus V(f) = \{\mathfrak{p} \in \operatorname{Spec} A : f \notin \mathfrak{p}\}.$$

Lemma 3.6. $\{D(f)\}_{f \in A}$ forms a basis for the Zariski topology on $\operatorname{Spec} A$. And $D(f) \cap D(g) = D(fg)$, for any $f, g \in A$.

Proof. Let $U = \operatorname{Spec} A \setminus V(\mathfrak{a})$. Then $U = \operatorname{Spec} A \setminus \bigcap_{f \in \mathfrak{a}} V(f) = \bigcup_{f \in \mathfrak{a}} D(f)$. To prove the second statement, notice that $f, g \notin \mathfrak{p}$ if and only if $fg \notin \mathfrak{p}$ for any prime ideal $\mathfrak{p}$. □

Sometimes we also write $X_f := D(f)$ for $X = \operatorname{Spec} A$. This is because $D(f)$ is isomorphic to $\operatorname{Spec} A_f$, the spectrum of the localization of A at f.

Corollary 3.7. [Ha, Exercise II.2.1] Let A be a ring, $f, g \in A$ be some elements, then the subspace $D(f)$ is homeomorphic to $\operatorname{Spec} A_f$.

Proof. This is because $\operatorname{Spec} A_f = \{\mathfrak{p}A_f : f \notin \mathfrak{p}\}$. □

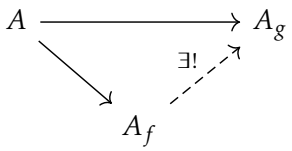
3.1.2 $\operatorname{Spec} A$ as a locally ringed space

Now we define the structure sheaf on $X = \operatorname{Spec} A$. We don't give the definition of the sheaf directly as [Ha]: define $\mathscr{O}_X(U)$ to be the set of functions $s : U \to \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}}$, such that $s(\mathfrak{p}) \in A_{\mathfrak{p}}$ and s is locally a quotient of elements in A.

Instead, we use the construction in Proposition 2.9, to define the base sheaf on distinguished open sets $\{D(f)\}_{f \in A}$, and glue it to obtain the sheaf on X.

By Corollary 3.7 and [Ha, Theorem I.3.2], it's natural to define $\mathscr{O}_X(D(f)) := A_f$. If $D(g) \subseteq D(f)$, then $g \in (f)$. And there is a natural map $A_f \to A_g$ given

by Lemma 3.8, since if g is invertible, then f is invertible (write $g = uf$, then $f(ug^{-1}) = 1$)



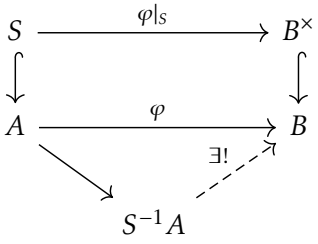
So we define the restriction map $\mathrm{res}_{D(f),D(g)}$ to be the natural map $A_f \to A_g$. The functoriality is also induced from Lemma 3.8.

Lemma 3.8. [Sta, Lemma 07K0] Let A be a ring, S be a multiplicative subset of A. Let M, N be A-modules. Assume all the elements of S act as automorphisms on N. Then the canonical map

$$\mathrm{Hom}_R(S^{-1}M, N) \to \mathrm{Hom}_R(M, N)$$

induced by the localization, is an isomorphism.

Let B be a ring, then the universal property of localization can be stated as follows. Each ring homomorphism $\varphi : A \to B$ such that $\varphi(S) \subseteq B^\times$, i.e., the image of each point $s \in S$ is invertible in B, factors uniquely through the localization map $A \to S^{-1}A$.



The next lemma is widely used in the proof of Proposition 3.10 and moreover Proposition 9.1. Serre has described this as a "partition of unity", according to Lemma 1.38 in differential geometry. This lemma also proves that $X = \mathrm{Spec}\, A$ is quasi-compact, i.e., each open cover of X admits a finite sub open cover.

Lemma 3.9 (Partition of unity). [Ha, Exercise II.2.13(b)], [E&H, Lemma I-16] Let $X = \mathrm{Spec}\, A$, and let $\{f_\alpha\}$ be a collection of elements of A. Then the open sets $D(f_\alpha)$ cover X if and only if the elements generate the unit ideal.

In particular, X is quasi-compact.

Proof. If $X = \bigcup D(f_\alpha)$, then $V(\sum(f_\alpha)) = \bigcap V(f_\alpha) = \varnothing$, hence $1 \in \sum(f_\alpha)$, by Corollary 3.4. Therefore, there are finitely some f_i and $e_i \in A$ such that $1 = \sum_{i=1}^n e_i f_i$. □

Proposition 3.10. [E&H, Lemma I-17], [Va, Theorem 4.1.2] Let $X = \operatorname{Spec} A$, and suppose that $D(f)$ is covered by open sets $D(f_i) \subseteq D(f)$.

(a) **Identity:** If $g \in A_f$ vanishes in each A_{f_i}, i.e., $[g]_{f_i} = 0$, then $g = 0$ in A_f.

(b) **Glueing:** If for each i there is $g_i \in A_{f_i}$ such that for each pair i, j, $[g_i]_{f_i f_j} = [g_j]_{f_i f_j}$, then there is an element $g \in A_f$ such that $[g]_{f_i} = g_i$ in A_{f_i}.

Equivalently, if $\mathcal{B} = \{D(f)\}_{f \in A}$, then $\mathscr{O}_X(D(f)) = A_f$ gives a $\mathcal{B}$-sheaf on X, hence determines a sheaf on $\operatorname{Spec} A$, by Proposition 2.9.

Proof. We first assume that $f = 1$, then we can assume that $X = \bigcup_{i=1}^n D(f_i)$, by Lemma 3.9.

(a) If $[g]_{f_i} = 0$, then there is $n_i \in \mathbb{N}$ such that $f_i^{n_i} g = 0$, for $i = 1, \dots, n$. Therefore, we can choose a sufficiently large N, such that $f_i^N g = 0$ for each i. By the partition of unity, $1 = \sum a_i f_i \in (f_1, \dots, f_n)$, and

$$1 \in (\sum a_i f_i)^{nN} \in (f_1^N, \dots, f_n^N),$$

hence $1 = \sum e_i f_i^N$. Therefore,

$$g = (\sum e_i f_i^N) g = \sum e_i (f_i^N g) = 0.$$

(b) Let $g_i = h_i / f_i^{n_i}$ in A_{f_i} for some $h_i \in A$, then we can assume that $f_i^{n_i} g_i = h_i$ in A_{f_i}. Similarly, we can take a sufficiently large N such that $f_i^N g_i = h_i$ for each $i = 1, \dots, n$. Now the compatible condition implies that

$$f_i^N h_j = f_i^N f_j^N g_i = f_i^N f_j^N g_j = f_j^N h_i \quad \text{in } A_{f_i f_j}.$$

By partition of unity, write $1 = \sum e_i f_i^N$. Define $g = \sum e_i h_i$. Then

$$f_i^N g = f_i^N \sum e_j h_j = \sum e_j (f_i^N h_j) = \sum e_j f_j^N h_i = h_i = f_i^N g_i$$

which implies that $[g]_{f_i} = g_i$ in A_{f_i}. This proves the glueing axiom.

Returning to the case of arbitrary f. Set $X' = D(f) \cong \operatorname{Spec} A_f$, and $f_i' = f f_i$, then $X'_{f_i'} = X_{f_i}$ and $X'_f = \operatorname{Spec} A_f$, so we can apply the case already proved to the primed data. □

By the terminology in quasi-coherent sheaves, we sometimes write $\widetilde{A}$ for the structure sheaf of $\operatorname{Spec} A$.

Corollary 3.11. [Ha, Proposition II.2.2] Let A be a ring. Let $X = \operatorname{Spec} A$, let $\mathscr{O}_X$ be the structure sheaf on X.

1. For any $f \in A$, $\mathscr{O}_X(D(f)) = A_f$. In particular, $\Gamma(X, \mathscr{O}_X) = A$.
2. For any prime ideal $\mathfrak{p}$, the stalks $\mathscr{O}_{X,[\mathfrak{p}]} = A_{\mathfrak{p}}$.
3. $\operatorname{Spec} A$ is a locally ringed space.

Proof. To prove (b), it suffices to prove $\varinjlim_{f \notin \mathfrak{p}} A_f = A_{\mathfrak{p}}$. Since for $f \notin \mathfrak{p}$, f is invertible in $A_{\mathfrak{p}}$, there is a natural map $A_f \to A_{\mathfrak{p}}$. The compatibility is also given by the universal property of localization Lemma 3.8.

Now suppose there is a ring B with compatible homomorphisms $\varphi_f : A_f \to B$. Notice that $1 \notin \mathfrak{p}$, hence $\varphi := \varphi_1 : A \to B$ sends each $f \notin \mathfrak{p}$ to a unit $\varphi(f) = \varphi_f(f) \in B^\times$. Therefore, φ factors uniquely through $A \to A_{\mathfrak{p}}$. This proves that $\varinjlim_{f \notin \mathfrak{p}} A_f = A_{\mathfrak{p}}$. □

3.2 Morphisms To Affine Schemes

Let $(X, \mathscr{O}_X)$ be a locally ringed space. Now we describe the morphism $X \to \operatorname{Spec} A$.

Lemma 3.12. [Sta, Lemma 01HZ] Let $(X, \mathscr{O}_X)$ be a locally ringed space, $s \in \Gamma(X, \mathscr{O}_X)$ a global section, let

$$D'(s) := \{P \in X : s_P \notin \mathfrak{m}_{X,P} \subseteq \mathscr{O}_{X,P}\} = \{P \in X : s_P \neq 0 \in \mathscr{O}_{X,P}/\mathfrak{m}_{X,P}\}$$

be the support of s. Then

(a) $D'(s)$ is an open subset.

(b) Moreover, $s|_{D'(s)}$ is invertible in $\mathscr{O}_X(D'(s))$.

(c) If $X = \operatorname{Spec} A$, then $D'(s) = D(s)$.

Proof. (a) Since for every point $P \in D'(s)$, s is invertible in the stalk $\mathscr{O}_{X,P}$, there is an open neighborhood of V_P, with a section $t_P \in \mathscr{O}_X(V_P)$, such that $(t_P)(s|_{V_P}) = 1$. Now we claim that $V_P \subseteq D'(s)$. Indeed, for any $Q \in V_P$, V_P is an open neighborhood of Q, and $s|_{V_P}$ is invertible, thus $Q \in D'(s)$, and $D'(s) = \bigcup_{P \in D'(s)} V_P$ is an open subset.

(b) Therefore, $\{V_P\}$ forms an open cover of $D'(s)$, and since $(t_P|_{V_P \cap V_Q})(s|_{V_P \cap V_Q}) = ((t_P)(s|_{V_P}))|_{V_P \cap V_Q} = 1$, we have $t_P|_{V_P \cap V_Q} = (s|_{V_P \cap V_Q})^{-1} = t_Q|_{V_P \cap V_Q}$. We can obtain the section $t \in \mathscr{O}_X(D'(s))$ by glueing all the t_P. We have $t(s|_{D'(s)}) = 1$, since $(t_P)(s|_{V_P}) = 1$ for every $P \in D'(s)$. Thus, $s|_{V_P}$ is invertible in $\mathscr{O}_X(D'(s))$.

(c) Notice that for $f \in A$, $D(f) = \{\mathfrak{p} \in \operatorname{Spec} A : [f]_{\mathfrak{p}} \text{ is invertible in } A_f\}$, by Lemma 3.8. □

Theorem 3.13. [E&H, Theorem I-40] For any locally ringed space X and any ring A, the morphisms

$$(f, f^\sharp) : X \to \operatorname{Spec} A$$

are in 1-1 correspondence with the homomorphisms of rings

$$\varphi : A \to \Gamma(X, \mathscr{O}_X).$$

Proof. If $(f, f^\sharp)$ is given, then φ is defined as $f^\#(\operatorname{Spec} A) : A \to \Gamma(X, \mathscr{O}_X)$. Now for a given ring homomorphism $\varphi : A \to \Gamma(X, \mathscr{O}_X)$, we define $f : X \to \operatorname{Spec} A$ as follows. For any point $x \in X$, let $\mathfrak{m}_{X,x} \subseteq \mathscr{O}_{X,x}$ be the maximal ideal in the stalks. Let φ_x be the composition of $\varphi : A \to \Gamma(X, \mathscr{O}_X)$ and the localizing map $\Gamma(X, \mathscr{O}_X) \to \mathscr{O}_{X,x}$. Then $\mathfrak{p}_x := \varphi_x^{-1}(\mathfrak{m}_{X,x})$ is a prime ideal in A. The morphism is given by

$$f : |X| \to |\operatorname{Spec} A|, \quad x \mapsto [\mathfrak{p}_x] = \varphi_x^{-1}(\mathfrak{m}_{X,x}).$$

f is continuous. Let $D(g)$ be a distinguished open set in $\operatorname{Spec} A$, for some $g \in A$. Then

$$f^{-1}(D(g)) = \{x \in X : g \notin \varphi_x^{-1}(\mathfrak{m}_{X,x})\} = \{x \in X : [\varphi(g)]_x \notin \mathfrak{m}_{X,x}\} = D'(\varphi(g))$$

is an open set in X, by the lemma above.

$f^\sharp$ is defined locally to be the composition

$$f^\sharp(D(g)) : A_g \xrightarrow{\varphi_g} \Gamma(X, \mathscr{O}_X)_{\varphi(g)} \to \mathscr{O}_X(D'(\varphi(g)))$$

where the map $\Gamma(X, \mathscr{O}_X)_{\varphi(g)} \to \mathscr{O}_X(D'(\varphi(g)))$ is induced from the universal property of localization, and that $\varphi(g)|_{D'(\varphi(g))}$ is invertible in $\mathscr{O}_X(D'(\varphi(g)))$. □

Corollary 3.14. [Ha, Exercise II.2.5] $\operatorname{Spec} \mathbb{Z}$ is the final object in the category of locally ringed spaces, thus the final object in the category of schemes, given by the structure map $\mathbb{Z} \to \Gamma(X, \mathscr{O}_X)$ for any locally ringed space $(X, \mathscr{O}_X)$.

Theorem 3.15. [Ha, Exercise II.2.4] Spec can be viewed as a contravariant functor $\mathsf{CRing} \to \mathsf{LRS}$, and the global section functor Γ is a contravariant functor $\mathsf{LRS} \to \mathsf{CRing}$. Then $(\operatorname{Spec}, \Gamma)$ adjoint.

Corollary 3.16. [Ha, Proposition II.2.3]

3.3 Schemes

Definition 3.17 (Schemes)**.**

Definition 3.18 (Schemes over a base)**.**

Lemma 3.19 (Open subschemes)**.** [Ha, Exercise II.2.2]

Definition 3.20 (Residue field)**.**

Lemma 3.21. [Ha, Exercise II.2.7]

Definition 3.22. If X is a topological space, and $Z \subseteq X$ an irreducible closed subset of X, a *generic point* for Z is a point $\zeta \in Z$ such that $Z = \overline{\{\zeta\}}$.

Given two points x, y of a topological space X, we say that x is a *specialization* of y, and y is a *generization* of x, if $x \in \overline{\{y\}}$.

For example, if the nilradical $\mathrm{nil}(A) := \{f \in A : f^n = 0 \text{ for some } n \in \mathbb{N}\}$ is a prime ideal, then by Lemma 3.2 , $[\mathrm{nil}(A)]$ is a generic point in $\operatorname{Spec} A$, since $[\mathrm{nil}(A)]$ is equal to the radical ideal of the zero ideal (0). In particular, if A is an integral domain, then $[(0)]$ is a generic point for $\operatorname{Spec} A$.

Proposition 3.23. [Ha, Exercise II.2.9] If X is a scheme, then any irreducible closed subset has a unique generic point.

3.4 Examples of Affine Schemes

Example 3.24 (Affine line)**.** [Ha, Example II.2.3.3, Exercise II.2.10]

Example 3.25 (Affine plane)**.** [Ha, Example II.2.3.4]

Example 3.26 (Affine line in characteristic p)**.** [Ha, Exercise II.2.11]

Recall that a ring A is reduced if it contains no nontrivial nilpotents, i.e., for $f \in A$, $f^n = 0$ for some $n \in \mathbb{N}$ if and only if $f = 0$. It's easily to prove that, $A/\mathfrak{a}$ is reduced if and only if $\mathfrak{a}$ is a radical ideal, i.e., $\sqrt{\mathfrak{a}} = \mathfrak{a}$.

$A/\mathfrak{a}$ is	reduced	an integral domain	a field
$\mathfrak{a}$ is	radical	prime	maximal

Example 3.27 (Reduced affine scheme)**.** Let k be a field. Then $\operatorname{Spec} k[x]/(x) \cong \operatorname{Spec} k$ is a singleton as a topological space. Notice that $|\operatorname{Spec} k[x]/(x^2)|$ consists of a unique point $[(x)]$, thus $|\operatorname{Spec} k[x]/(x^2)| = |\operatorname{Spec} k[x]/(x)|$. But the function ring k and $k[x]/(x^2)$, while $k[x]/(x^2)$ is nonreduced.

Generally, let A be a ring, $\operatorname{nil}(A)$ be the nilradical ideal of A. Then the inclusion

$$\operatorname{Spec} A/\operatorname{nil}(A) \to \operatorname{Spec} A$$

is a bijection, by Lemma 3.2. We denote $(\operatorname{Spec} A)_{\mathrm{red}} := \operatorname{Spec} A/\operatorname{nil}(A)$, and call it the *reduced scheme associated to* $\operatorname{Spec} A$. As the underlying sets (and also topological space), $(\operatorname{Spec} A)_{\mathrm{red}}$ and $\operatorname{Spec} A$ are the same.

Geometrically, a function on $\operatorname{Spec} A$ vanishes at every point if and only if it is nilpotent. Therefore, there are indeed some functions which are not determined by their values in the nonreduced schemes, they are the nilpotents vanishing at every point but not equal 0. For example, $x+(x^2)$ vanishes at $[(x)]$, but x is not zero in $k[x]/(x^2)$.

Example 3.28 (Closed subscheme of affine scheme)**.** Let $X = \operatorname{Spec} A$ is an affine scheme. Then for each ideal $\mathfrak{a} \subseteq A$, the closed subset $V(\mathfrak{a})$ is endowed with a closed subscheme structure $\operatorname{Spec} A/\mathfrak{a}$. However, we can define different sheaf on $V(\mathfrak{a})$ to make it a closed subscheme. This is because $V(\mathfrak{a}) = V(\mathfrak{b})$ if and only if $\sqrt{\mathfrak{a}} = \sqrt{\mathfrak{b}}$, by Corollary 3.4.

Among all the closed subscheme structures defined on $V(\mathfrak{a})$, there is a smallest one, named *reduced induced closed subscheme structure*, defined to be $\operatorname{Spec} A/\sqrt{\mathfrak{a}}$.

We will prove later each closed subscheme of $\operatorname{Spec} A$ is of the form $\operatorname{Spec} A/\mathfrak{a}$.

3.5 Examples of Schemes

Example 3.29 (Zariski tangent space)**.** [Ha, Exercise II.2.8]

Example 3.30 (Glueing schemes). [Ha, Example II.2.3.5]

Example 3.31 (Double points line). [Ha, Example II.2.3.6]

4 From Varieties To Schemes

4.1 Proj Construction

Definition 4.1 (Proj).

Lemma 4.2. [Ha, Lemma II.2.4]

Corollary 4.3. [Ha, Proposition II.2.5]

Example 4.4. [Ha, Example II.2.5.1]

4.2 From Varieties To Schemes

Proposition 4.5. [Ha, Proposition II.2.6]

5 Some Properties of Schemes

5.1 Toplogical properties

5.2 Reducedness & Intergrality

Proposition 5.1. [Ha, Exercise II.2.3]

5.3 Noetherian Schemes

6 First Properties of Morphisms

6.1 Finiteness

6.2 Immersions

6.3 Fibred Products

7 Separatedness & Properness

7.1 Separatedness

7.2 Properness

7.3 Projective Morphisms

8 Sheaves of Modules

8.1 More Operations on Sheaves

8.2 Invertible Sheaves

8.3 Locally Free Sheaves & Vector Bundles

9 Quasi-coherent Sheaves

Proposition 9.1. [Ha, Proposition II.5.6] Let X be an affine scheme, let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence of $\mathscr{O}_X$-modules, and assume that $\mathscr{F}'$ is quasi-coherent. Then the sequence

$$0 \to \Gamma(X, \mathscr{F}') \to \Gamma(X, \mathscr{F}) \to \Gamma(X, \mathscr{F}'') \to 0$$

is exact.

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